

# Human vs. AI in Symbolic Music: An Online Listening Study on Bach-Style Chorales

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## Introduction & Method

We conducted a music perception study to evaluate human vs. AI discrimination in Bach-style chorales. Participants ( $N = 295$ ) listened to **20 isolated musical phrases** and categorized each as **AI-** or **human-composed**. AI snippets were generated using *NotaGen* [1], a state-of-the-art symbolic music generation model. All musical snippets (both human and AI) were manually curated. Participation was open to the public with **no prior music qualification required**. Prior to and following the experiment, we collected metadata such as age, musical background, and self-reported confidence levels in order to better contextualize participant performance.

## Experiment Design

Pre-Survey → 20 Listening Trials → Post-Survey  
(Order randomized)

### Stimuli

- **Balanced Set:** 10 Bach chorale phrases vs. 10 *NotaGen* phrases
- **Audio Uniformity:** All excerpts rendered with an identical MIDI harpsichord soundfont
- **Selection:** All snippets were manually curated

### Measured Parameters

- **Demographics:** Age, weekly AI usage (Yes/No).
- **Musical Background:** Plays instrument (Yes/No), sings in choir (Yes/No), self-reported experience (4-point Likert scale).
- **Confidence:** Pre- and post-experiment self-ratings (4-point Likert scale).

## Qualitative Analysis

Ricarda...

## Age Distribution

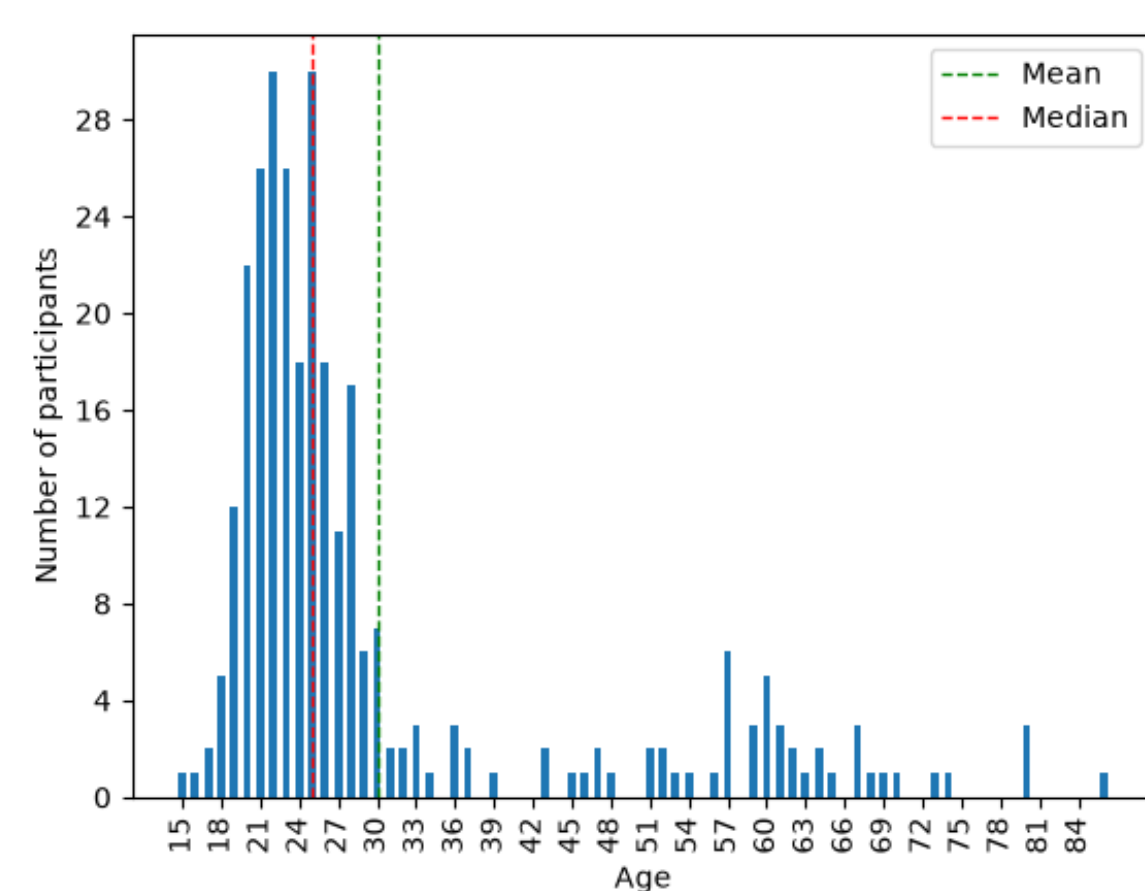


Figure 1: Participant age distribution ( $N = 295$ )

## Performance by Choir Experience

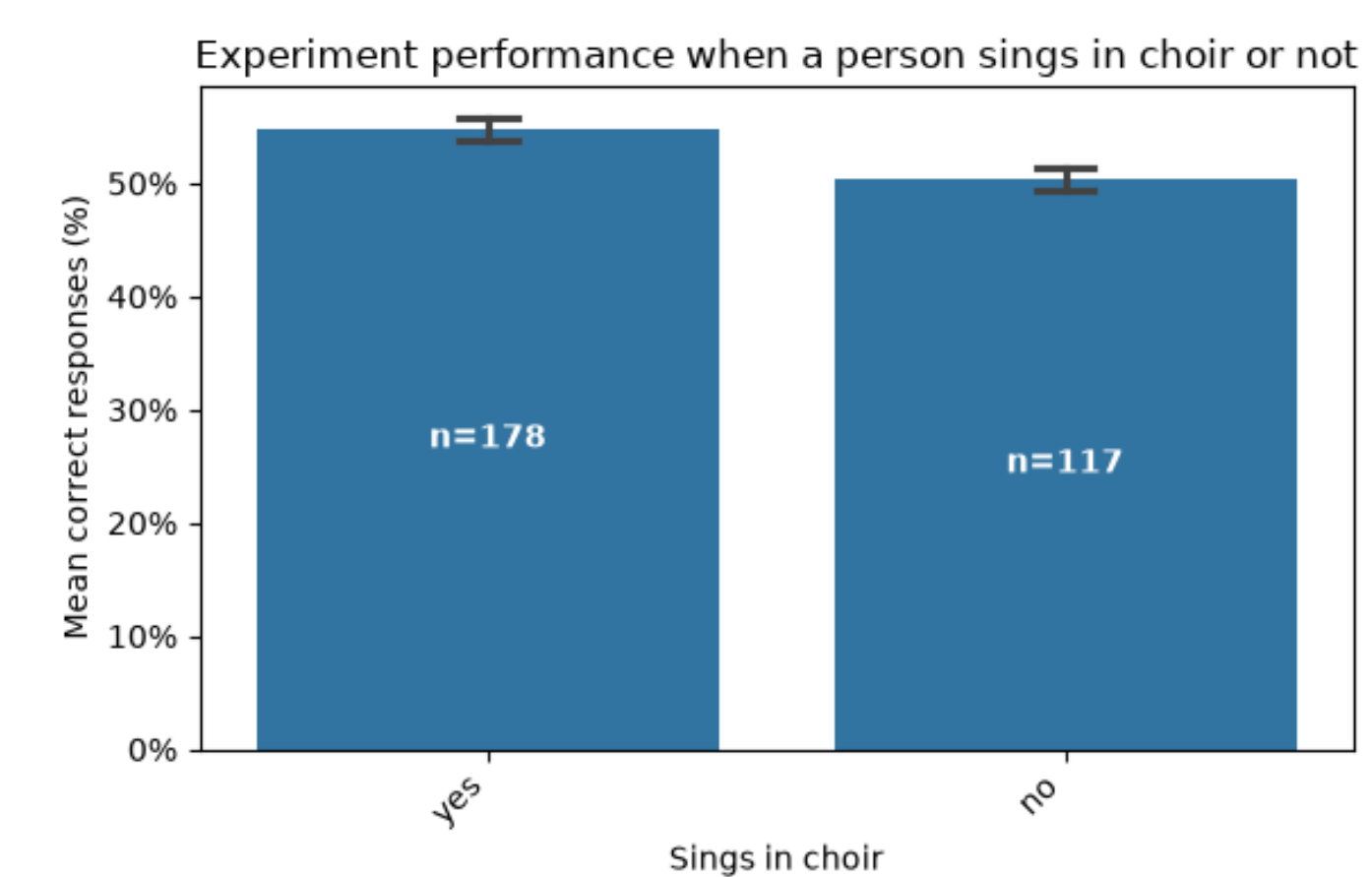


Figure 3: Accuracy comparison by participants' choir experience

## Performance by Musical Experience

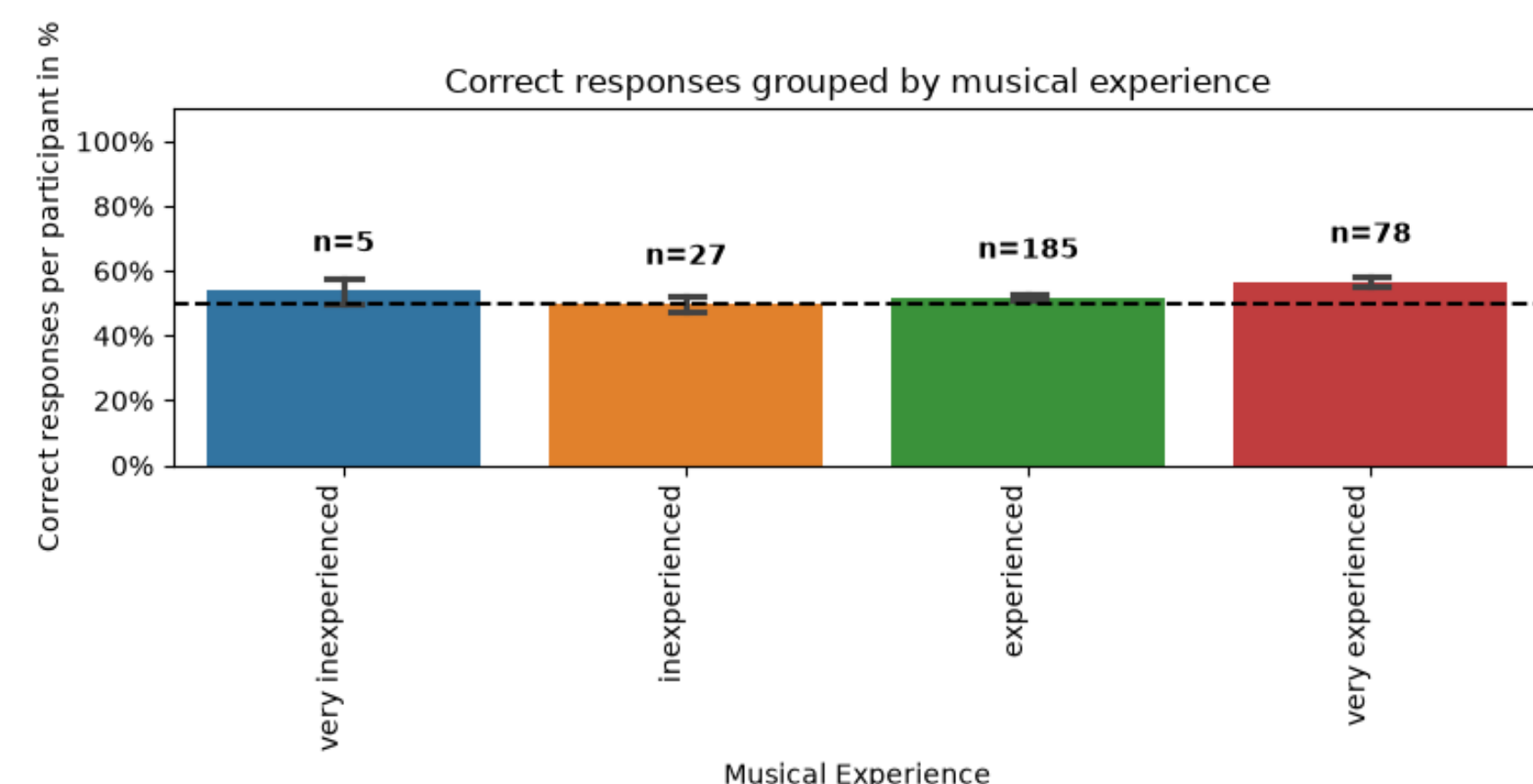


Figure 2: Discrimination accuracy across participants' prior musical experience

## Confidence Shifts

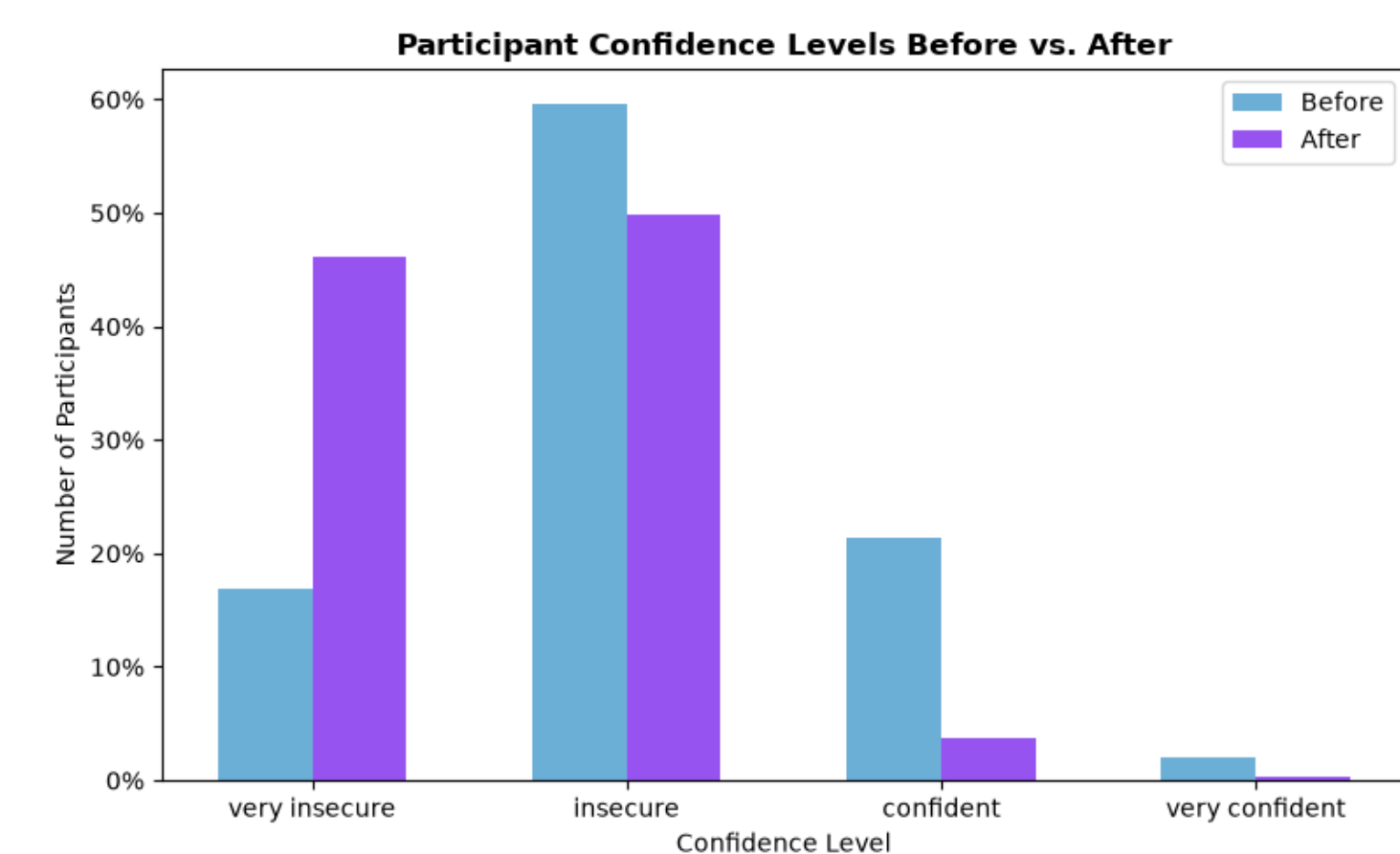


Figure 4: Self-reported confidence pre- vs. post-trial.

## Overview: Performance Per Participant

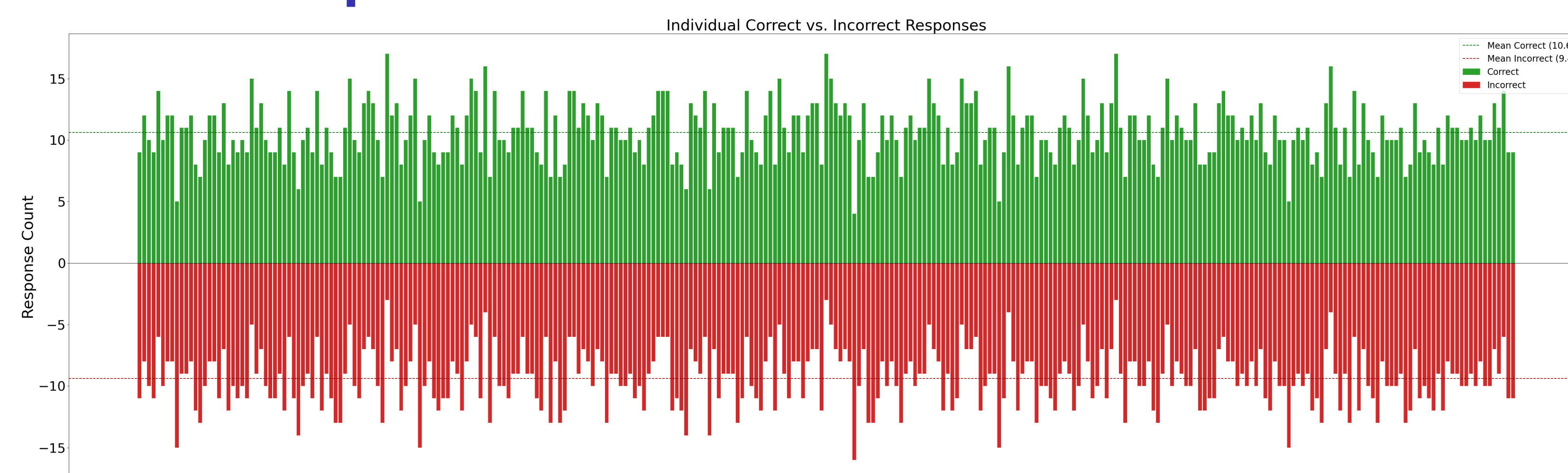


Figure 5: Individual participant accuracy scores across all trials.

## References

- [1] Yashan Wang et al. *NotaGen: Advancing Musicality in Symbolic Music Generation with Large Language Model Training Paradigms*. Mar. 2025. DOI: 10.48550/arXiv.2502.18008. arXiv: 2502.18008 [cs]. URL: <http://arxiv.org/abs/2502.18008> (visited on 01/21/2026).

## Try the Experiment



Can you distinguish AI from human?

Scan to test your skills in our study:

[jasonvla.github.io/ai-chorales\\_website](https://jasonvla.github.io/ai-chorales_website)